Competency Level 9.9: Uses sub-programs in programming

Learning Outcomes:

- Briefly describes the functions
- Lists and briefly describes the types of functions
- Identifies the structure of a function
- Compares local and global variables
- Identifies the behavior of a variable in terms of life time
- Identifies the need of return values and writes functions to obtain the appropriate return value
- Writes functions using relevant parameters and arguments
- Uses user defined functions

Function

A function is a block of code which only runs when it is called.
You can pass data, known as parameters, into a function.
So the function can return data as a result.

Creating Functions in Python
In Python a function is defined using the **def** keyword:

Example:

```
def my_function():
    print("This is my first function")
```

Calling a Function
To call a function, use the function name followed by parenthesis:

Example 20:
CodeOutput

```
File Edit Format Run Options Windows Help
def my_function():
  print("This is my first function")

my_function()
```

```
>>>
This is my first function
>>>
```

Parameters
Information can be passed to functions as a parameter.
Parameters are specified after the function name, inside the parentheses. You can add as many parameters as you want, just separate them with a comma.

The following example has a function with **one parameter (fname)**. When the function is called, we pass along a first name, which is used inside the function to print the full name:

Example 21:

Code

```
def family_name():
      fname=input("Enter Your Name :")
      print(fname + " De Silva")
```

Output

```
Python 3.3.1 (v3.3.1:d9893d13c628, Apr  6 2013, 20:25:12) [MSC v.1600 32 bit (
Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>> ================================ RESTART ================================
>>>
>>> ================================ RESTART ================================
>>>
>>> family_name()
Enter Your Name :Kamal
Kamal De Silva
>>>
```

Default Parameter Value

The following example shows how to use a default parameter value.
If we call the function without parameter, it uses the default value:

Example 22:

Code

```
def add(a,b):
  c=a+b
  print("a+b= ",c)
```

Output

```
Python 3.3.1 (v3.3.1:d9893d13c628, Apr  6 2013, 20:25:12) [MSC v.1600 32 bit (
Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>> ================================ RESTART ================================
>>>
>>> add(3,8)
a+b= 11
>>>
```

Return Values

To let a function return a value, use the return statement:

Example 23:

Code Output

```
def return_val(a):
  return a * 2

print(return_val(4))
print(return_val(2))
print(return_val(6))
```

```
>>>
8
4
12
>>>
```

Scope of Variables

All variables in a program may not be accessible at all locations in that program. This depends on where you have declared a variable.
The scope of a variable determines the portion of the program where you can access a particular identifier.

There are two basic scopes of variables in Python:

- Global variables
- Local variables

Global vs. Local variables

Variables that are defined inside a function body have a local scope, and those defined outside have a global scope.
This means that local variables can be accessed only inside the function in which they are declared, whereas the global variables can be accessed throughout the program body by all functions. When you call a function, the variables declared inside it are brought into scope.

Example 24:
Code

```
File  Edit  Format  Run  Options  Window  Help
r=float(input("Enter the radius value in cm "))
pi=3.14
def circle_area():
    area=pi*r*r
    print("Area of the circle for radius", r,"  : ", area, "square cm")
def circle_circumference():
    circum=2*pi*r
    print("circumference of the circle for radius ", r," cm : ", circum, "cm")
```

Output

```
==================== RESTART: C:\Users\NIE\Desktop\glo.py ====================
Enter the radius value in cm 4
>>> circle_area()
Area of the circle for radius 4.0   :  50.24 square cm
>>> circle_circumference()
circumference of the circle for radius  4.0  cm :  25.12 cm
>>> |
```